Aparna Krishnan

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Education

Virginia Polytechnic Institute and State University, Blacksburg, VA **August 2023 – Present**
PhD. Theoretical Chemistry *Advisor : Prof T Daniel Crawford* *3.82/4.0*
Honors: Phi Kappa Phi
Experience: Time-dependent coupled cluster, Linear response theory, Spectroscopy, Diagonalization algorithms, Analytic gradient theory, Numerical differentiation, Scientific automation pipelines

Birla Institute of Technology and Science Pilani, Goa **August 2018 – May 2023**
Int. MSc.(Hons.) Chemistry and B.E.(Hons.) Mechanical Engineering *9.01/10.0*
Honors: Rank 1, Chemistry
Coursework: Optimization, Group theory, Statistical Mechanics, Quantum Mechanics, Quantum Chemistry and Spectroscopy, Electronic Structure Theory, Finite Element Methods, Numerical Methods for Software Mathematics, Machine Learning

Publications

†These authors contributed equally.

Chemical Interpretation of Time-Dependent Coupled-Cluster Theory
A. Krishnan, H. E. Kristiansen, B. G. Peyton, T. D. Crawford, T. B. Pedersen
*Journal of Physical Chemistry A - **accepted**. arXiv:2605.17409*

Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism within Configuration Interaction Theory
A. Krishnan, B. Shumberger, T. D. Crawford
Journal of Chemical Theory and Computation - submitting 2026

Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism at the Coupled-Cluster Singles and Doubles Level
A. Krishnan, T. D. Crawford
In preparation

Investigating Correlation in Analytic Vibrational Circular Dichroism: HF, MP2, CISD, and CCSD
A. Krishnan, T. D. Crawford
In preparation

Real-Time Unitary Coupled-Cluster Methods for Molecular Spectroscopy
A. Melekamburath†, **A. Krishnan**†, E. F. Valeev, T. D. Crawford
In preparation

PyCC: An Open-Source Python-Based Coupled Cluster Package
A. Krishnan, T. D. Crawford, et al.
In preparation

Deformation Characterization of Additive Manufactured Polyether Ether Ketone Using Mirror Assisted Image Correlation
V. Gupta, **A. Krishnan**, N. I. Thiruselvam, D. M. Kulkarni, V. V. Chaudhari, S. Suraj, R. Das
Submitted

Research Experience

Virginia Polytechnic Institute and State University **August 2023 – Present**
Graduate Research Assistant, Advisor: Prof T Daniel Crawford *Blacksburg, USA*

- Developed analytic vibrational circular dichroism theory at the CISD, MP2, and CCSD levels via a density-matrix reformulation of atomic axial tensors; proved a 2-RDM vanishing theorem for the orbital–orbital AAT block that removes an entire class of terms.
- Implemented complete analytic pipelines (Hessians, atomic polar tensors, atomic axial tensors, rotational strengths), validated against finite difference to better than 10^{-7}.

- Showed that real-time and equation-of-motion unitary coupled-cluster excitation energies are not equivalent at finite BCH truncation; derived the discrepancy in closed form and certified it against exact operator algebra in an exactly solvable model.
- Contributor to PyCC (16 merged commits): implemented TD-EOM-CC methods, analytic CISD-VCD APTs/AATs/Hessians, CISD polarizabilities via 2n+1 and explicit routes, and orbital decomposition analysis.

Virginia Polytechnic Institute and State University **August 2022 – May 2023**
Thesis Intern, Advisor: Prof T Daniel Crawford *Blacksburg, USA*

- Automated the spin integration of coupled cluster equations and generation of code, reducing manual calculation time by 80 percent and allowing research team to focus on complex quantum chemistry challenges.
- Formulated a multi-scale approach, combining MD and DFT, and devised a Python script that automated MD frame sampling; enhanced Raman spectra generation, and solvent shell definition, leading to a significant reduction in runtime from 10 days to 12 hours.

Indian Institute of Technology, Bombay **May 2021 – March 2022**
Research Intern, Advisor: Prof Rahul Maitra *Mumbai, India*

- Established a rigorous theoretical understanding of CC theory and explored the applications in quantum chemistry.
- Developed an innovative engine that automates Wick's theorem. Also implemented a fully automatic equation and code generator for dual exponent formulation of CC theory involving scattering-type operators with minimal user input.

Dhio Research and Engineering Pvt Ltd. **May - July 2020**
Research Intern (Industry), Advisor: Santosh NL *Bangalore, India*

- Simulated the visco-elastic properties of polymeric materials using J-OCTA and proposed processing and performance improvements for rubber-type polymers in the automotive industry; identified critical polymer architectures for better on-road tire performance, with ab-initio calculations 5x faster than the existing baseline.

Software

vcd-kit: analytic vibrational circular dichroism at the HF, MP2, CISD, and CCSD levels - atomic axial tensors, polar tensors, Hessians, and rotational strengths from a density-matrix formulation.
PyCC: open-source Python coupled-cluster package. Contributor (16 merged commits) - TD-EOM-CC, VCD pipelines, orbital decomposition analysis.
Biorthogonal Jacobi-Davidson: non-Hermitian eigensolver tracking left and right eigenvectors with enforced biorthogonality.
CC Equation Generator: symbolic derivation of dual-exponent CC equations with automatic `np.einsum` code generation.
Solvated Spectra Pipeline: automated MD-to-DFT workflow for chiroptical spectra of solvated molecules; 10 days to 12 hours.

Projects

Density-based reformulation of correlated vibrational circular dichroism

- Originated a density-matrix reformulation of atomic axial tensors (AATs) after recognising that machinery developed for the diagonal Born-Oppenheimer correction could be transferred to chiroptical response, and built it into a single framework spanning HF, MP2, CISD, and CCSD.
- Proved a 2-RDM vanishing theorem for the orbital-orbital AAT block, removing an entire class of terms and making the framework cleanly extensible to coupled cluster.
- Implemented the complete analytic pipeline at each level - Hessians, normal-mode analysis, atomic polar tensors, atomic axial tensors, rotational strengths - validated against finite difference to better than 10^{-7}. Used the Z-vector/Lagrangian approach at MP2 to avoid solving coupled-perturbed equations per nuclear displacement.
- Extended the framework with gauge-including atomic orbitals (GIAOs) for origin-independent AATs, improving accuracy for basis-set-dependent magnetic response properties.

Real-time coupled cluster methods for the calculation of field-dependent molecular properties

- Revealed the orbital and symmetry contribution to peaks in an absorption spectra. Crafted an algorithm using autocorrelation as a sole metric for deriving field-dependent properties and orbital decomposition, which can reduce storage requirements by 95 percent.

Structural non-equivalence of real-time and EOM unitary coupled cluster

- Showed that the RT/EOM equivalence guaranteed in conventional coupled cluster fails for the unitary ansatz at finite BCH truncation; derived the discrepancy in closed form, proved it vanishes in the two limits recovering the standard result, and certified it against brute-force operator algebra to 10^{-12}.

A bi-orthogonal Jacobi-Davidson method to solve the eigenvectors of a non-Hermitian system

- Engineered a bi-orthogonal Jacobi-Davidson (biJD) framework that maintains bi-orthogonality between left and right eigenvectors throughout, converging roughly 25x faster than a conventional Jacobi-Davidson implementation on the same test problem.

Mirror-assisted Digital Image Correlation (DIC) to extract side surface information

- Designed and validated a mirror-assisted 3D-DIC technique for measuring side surface deformations in 3D printed carbon-reinforced PEEK, enhancing accuracy and reducing equipment complexity.

Skills

Languages: Python, C++
Scientific Computing: NumPy, SciPy, SymPy (secondquant), einsum, h5py/HDF5, Pandas, Matplotlib, Jupyter
Quantum Chemistry: Psi4, Gaussian 16, CFOUR, PyCC
HPC & Tooling: SLURM, ARC cluster (Virginia Tech), Linux/Bash, Git/GitHub, LaTeX
Quantum Computing: Qiskit, Cirq, QDK, quantum algorithms
Machine Learning: PyTorch, TensorFlow, Keras, NLP
Certifications: Quantum Computing (QWorld), Machine Learning (Stanford), Quantum Excellence (QGSS'25), Quantum Error Correction (Google QAI, in progress)

Talks and Presentations

MES 2026, Greece - Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism within Correlated theories. *Upcoming, September 2026.*
SETCA 2026, Georgia State University, Atlanta - Density-matrix formulation of atomic axial tensors for CISD vibrational circular dichroism. **Best Poster Award.**
MRPSS 2025, Stockholm, Sweden - A tale of two decompositions: analyzing electronic excitations in real-time coupled-cluster.
SETCA 2025, University of Memphis, Tennessee - Orbital-resolved absorption spectra from RT-CC autocorrelation function.
SERMACS 2024, Atlanta, Georgia - Understanding real-time coupled-cluster absorption spectra: orbital contributions from individual transitions.
ESQC 2024, Palermo, Italy - Understanding real-time coupled-cluster absorption spectra: orbital contributions from individual transitions.
SETCA 2023, University of South Carolina, Columbia - Determination of optical properties by modeling complex solvent effects: molecular dynamics and density functional approach.

Achievements

Best Poster Award, SETCA 2026: Awarded for the density-matrix AAT formulation of CISD vibrational circular dichroism, Atlanta.
KVPY Scholar, 2019: Highly competitive national scholarship awarded by the Department of Science and Technology, Government of India, to students pursuing a career in science.
Elected member of Phi Kappa Phi: Elected for outstanding academic standing at Virginia Tech.
BITS Pilani Travel Award: Awarded to complete the Bachelor's thesis at a foreign university.